Psychosis & Schizophrenia

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- **Psychosis** is impaired reality testing. Patient can't differentiate his imagination from reality and thus his judgment is impaired. It includes negative and positive symptoms.

Negative symptoms	Positive symptoms
Symptoms that are missing (negative) in patient, But present in healthy people	Symptoms that are present (positive) in patient, But absent in healthy people
Alogia (lacks logic), Apathy (lacks emotion), Amotivation Ambivalence (lacks will), Asociability	Delusions: false fixed beliefs Hallucinations: perception of non existing stimuli

Schizophrenia is a chronic psychotic disorder, where **1 major or 2 minor** symptoms of the following occur for one month.

Major symptoms	minor symptoms
• **Running** auditory hallucinations, **Commenting** in **3rd** person	• delusions + hallucinations (that are neither delusions of control, nor 3rd person running commenting hallucinations)
• Delusions of **passivity** or thought **control**	• negative symptoms • Catatonic symptoms • Disorganized form of thinking (ideas are not related logically)

Prevalence: **1%,** equal in men & women,
peak age in men late teens, and in women thirties
more in low socio-economic classes

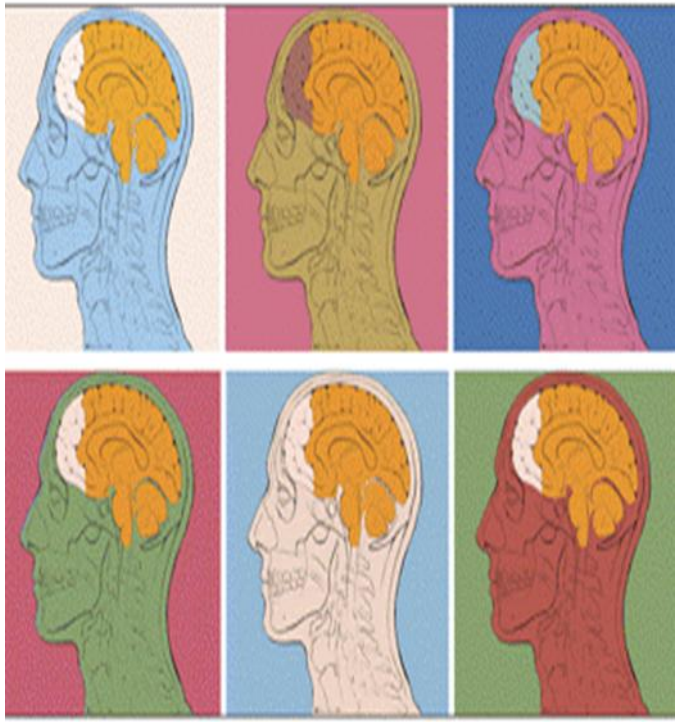
Etiology (nature & nurture)

Personal factors:

- Genetic predisposition (runs in families)
- Medical conditions:
  - **Chronic illnesses: porphyria**
  - **Medications: amphetamines & cannabis**
  - **Perinatal insults & head injury**

Social factors: Poor social support

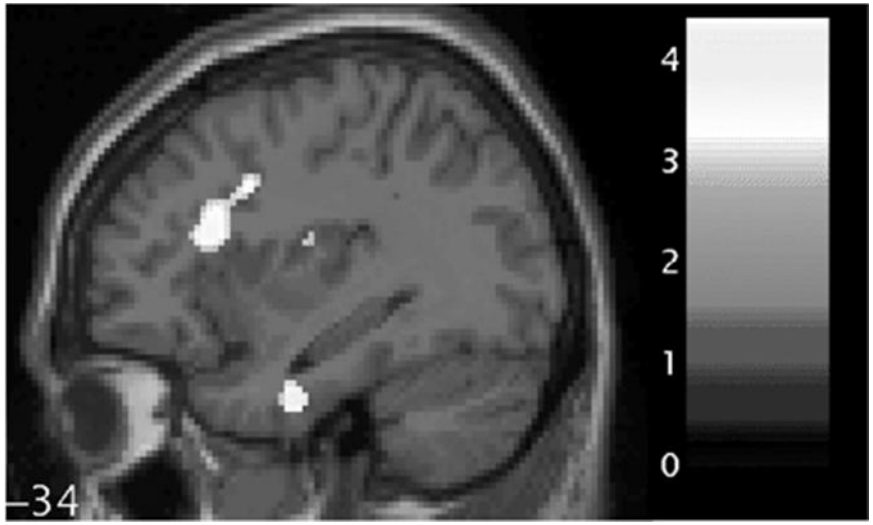
biology of psychosis: ***Alterations in CNS gene expression*** occur with normal growth and aging. Neuronal development and **dopamine** signaling show a ***failure of normal age-related down-regulation*** in subjects with schizophrenia.



- The illustration depicts differential age-related changes in prefrontal cortical gene expression in normal subjects (*top* row) versus those with schizophrenia (*bottom* row). ***(Torkamani & Thomas, 2010)***

There is growing evidence that schizophrenia is a ***disorder of cortical connectivity*** (**degenerative**). Specifically, fronto-temporal and fronto-parietal connections are thought to be functionally impaired. Diffusion tensor magnetic resonance imaging (DT—MRI) is a technique that has the potential to demonstrate structural disconnectivity in schizophrenia.

- Statistical parametric map (in sagittal section) showing regions of fractional anisotropy reduction (brightest areas) in the white matter of the left uncinate fasciculus and left arcuate fasciculus suggest that there is frontotemporal and frontoparietal structural disconnectivity in schizophrenia **(Burns et al, 2003)**



Complications

- Intangible emotional suffering
- Decreased functioning and lost opportunities
- Suicide
- Impulsivity and poor judgment render accidents, crimes and financial loses

Prognosis

- One third of patients reach complete remission.
- One third of patients reach partial remission.
- One third of patients remain as chronic cases.

Treatment

Where to treat?
hospitalize if there is risk (suicide – aggression – neglect -abuse) else treat in community
When? How long to treat?
guidelines recommend **one year** after symptom remission
Who to involve in treatment?
family and patient should be educated about the disorder
Why to treat? Goals:
- Response (symptoms decrease) Remission (symptoms vanish)
- Relapse (symptoms recur) and Recurrence prevention

What tools to use to treat? Strategies:
- **Psychotherapy: Cognitive remediation therapy**
- **Pharmacotherapy: Antipsychotics** +- augment with mood stabilizers in resistant cases and antidepressants in cases with depressive symptoms
- **Physical Therapy**: **Brain synchronization Therapy** (BST)

Antipsychotics

General practitioners (GPs) should avoid prescribing antipsychotics, better refer cases for specialist care due to serious side effects and intense follow up required.

Early diagnosis and referral of cases;

as well as good **psycho-education** for family

and good **rapport** with patient is the most essential role for a GP.

medication	mechanism	Side effect	adjuvant
1st generation e.g. chlorpromazine e.g. haloperidol	Anti-dopaminergic	Extra-pyramidal symptoms	Anti-mascarinic
2nd generation e.g. risperidone	Anti-dopaminergic Anti-serotonegic	Metabolic syndrome	Anti-diabetic
3rd generation e.g. aripiprazole	Dopamine modulator	akathesia	Beta blockers

Cognitive remediation therapy:

is a sort of cognitive rehabilitation, how to ignore hallucinations and distract oneself from them; also how to shake delusions

12 sessions (1/week), 1hour each

Brain synchronization therapy:

electrical current regulating the central nervous system when disrupted or "restarted" by induced seizure, brain circuits get to be more synchronized.

12 sessions (2/week), 0.8 amp for 6 sec. Bi-temporal under anesthesia

Other psychotic disorders

- **Delusional disorder:** the patient has delusions but still functioning. Subtypes include: grandiose delusions, persecution, jealousy, erotomania (delusions of love) and somatic delusions.
- **Schizoaffective disorder:** patient fulfills criteria of diagnosis of bipolar disorder and schizophrenia at the same time.
- **Acute psychotic episode:** psychotic features are present but in a duration less than 1 month.

Mood disorders

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Criteria to diagnose:

	• **Depressive episode 2 weeks** with at least 4 criteria of the following:	• **Manic episode 1 week** with at least 4 criteria:
1	Depressed mood	Elated mood
2	Decreased interest	Flights of ideas
3	Decreased energy	Hyper-talkative
4	Decreased Motor activity	Hyper-active
5	Decreased concentration	Easily distracted
6	Decreased Sleep satisfaction	Decreased need to sleep
7	Decreased **appetite**	Hyper-**sexual**
8	Decreased self esteem	Grandiose
9	Guilt	Hyper-sociable
10	suicide	Reckless spending, driving
	• Mixed episode 2 weeks of symptoms of both mania & depression	• **Hypomanic** episode **4 days** with 3 criteria only

Recurrent mood disorders

Recurrent depressive disorder (unipolar): episodes of depression recurring periodically	Bipolar I disorder: manic episodes alternating with depressive episodes
	Bipolar II disorder: hypomanic episodes alternating with depressive episodes

Persistent mood disorders

Dysthymia: depressive symptoms persisting for more than **2 years**	Hyper-thymia: manic symptoms persisting for more than **2 years**
	Cyclo-thymia: alternating symptoms persisting for more than **2 years** with no periods of normal mood in between

	depression	Mania
prevalence	10%	1%
Peak age	Early 30s	Late teens
sex	More in women	equal
class	low	high
➢ Medical causes	Malignancy Autoimmunity Organ failure hypothyroidism	Pellagra Multiple sclerosis Huntington disease
➢ Medications cause	Beta blockers Contraceptive pills	steroids

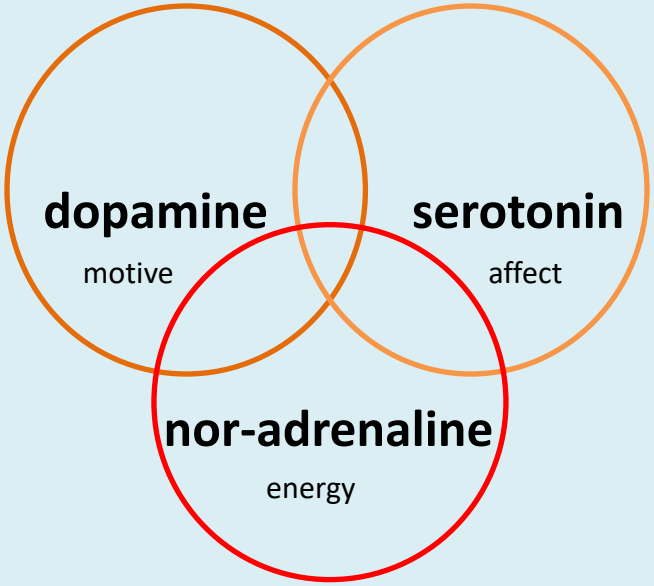
Etiology (nature & nurture)

Personal factors:

➢ Genetic predisposition (runs in families)

Social factors: Poor social support

Biology of depression
Tri-amine theory



Biology of mania
Kindling phenomena

- Neuronal hyper-excitability develops due to accumulating stress on susceptible genes **(Kindling).**
- **2nd messenger inside neurons are deregulated leading to abnormal activity and thus symptoms develop.**

Treatment

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hospitalize if there is risk (suicide – aggression – neglect -abuse) else treat in community
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What tools to use to treat? Strategies:
- **Psychotherapy: Cognitive behavioral therapy in unipolar & social rhythm therapy in bipolar**
- **Pharmacotherapy: Antidepressants in unipolar & mood stabilizers in bipolar**
- **Physical Therapy**: **Brain synchronization Therapy** (BST)

Antidepressants

General practitioners (GPs) could safely prescribing antidepressants, if they have comprehensive detailed knowledge about side effects, contra-indications and interactions

medication	mechanism	Side effect	interactions
SSRIs e.g. Fluoxetine	Selective serotonin Reuptake inhibitors	GIT upset, Anorgasmia, Bleeding tendency, **serotonin syndrome**	Cytochrome Enzyme **inhibitor**
SARI e.g. Trazodone	Serotonin agonist Reuptake inhibitor	**Priapism**, glucoma	SSRIs, SNRIs, NaSSA, Dnagents, aripirazole
SNRIs e.g. venlafaxine	Serotonin noradrinaline Reuptake inhibitors	**Hypertension**, **glucoma**	SSRIs, SARIs, NaSSA, DNagents, aripiprazole
NaSSA e.g. mirtazapine	Noradrenergic Specific serotonergic antidepressants	Increased appetite, increased sleepiness, **agranulocytosis**	SSRIs, SARIs, SNRIs, DNagents, aripirazole
DN agents e.g. bupropion	Dopaminergic Noradrenergic agents	**seizures**	SSRIs, SARIs, SNRIs, NaSSA, aripirazole

Mood stabilizers

General practitioners (GPs) should avoid prescribing mood stabilizers, better refer cases for specialist care due to serious side effects and intense follow up required.
Early diagnosis and referral of cases;
as well as good **psycho-education** for family
and good **rapport** with patient is the most essential role for a GP.

medication	mechanism	Side effect	interactions
Lithium	2nd messenger modulator	Hypothyroid, renal dysfn., **Neuroleptic malignant syndrome** (**cardiac** malformations if given in pregnancy)	SNRIs, SSRIs, aripiprazole, NSAIDs
Valproate	Na channel blocker GABAergic agent	Liver dysfn. (fetal valproate syndrome **FVS** if given in pregnancy)	SNRIs, SSRIs, Aripiprazole, asprin **Enzyme inhibitor**
Carbamazipine	Na channel blocker Serotonergic agent	Blood dyscrasias, arrythmia (**spina bifida** if given in pregnancy)	SNRIs, SSRIs, Aripiprazole, asprin **Enzyme inducer**

Cognitive behavioral therapy:

assumes that **cognitive errors** (perfectionism, black & white thinking, catastrophic thinking) contribute to depression. So detect and neutralize them.

12 sessions (1/week), 1hour each

Social rhythm therapy:

assumes that disruption of body rhythms and interpersonal conflicts cause kindling phenomena, so educate family and plan daily routine.

12 sessions (1/week), 1 hour each

Thank you